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# AI Project Proposal – Sidrit & Viola

# Topic: Intelligent Employee Attrition Prediction Agent

## Problem Definition

Employee attrition is a serious challenge for organizations because losing trained and experienced employees leads to increased recruitment costs, productivity loss, and reduced team stability. When a company hires an employee, it is important not only to assess current performance but also to understand whether the employee is likely to remain in the organization over time. If an employee leaves early, the organization must repeat the hiring and training process, which is costly and inefficient.

The goal of this project is to design an intelligent employee attrition prediction agent that predicts whether an employee is likely to leave the company based on demographic, job-related, and satisfaction-related features. This problem is formulated as a supervised learning binary classification task, where the target variable *Attrition* indicates whether an employee leaves the company (Yes) or stays (No). By identifying employees at risk of leaving in advance, the intelligent agent supports rational decision-making and enables organizations to take preventive actions to improve employee retention.

## Proposed AI Techniques

In this project, we use Random Forest and XGBoost as the two machine learning techniques for predicting employee attrition. Both methods are well suited for this problem because the dataset consists of structured employee data with multiple features that influence whether an employee stays or leaves.

- **Random Forest** is a machine learning algorithm that works by building many decision trees and combining their results to make a final prediction. Each tree learns different patterns from the data, and by averaging their predictions, the model becomes more stable and accurate. We chose Random Forest because it is easy to apply to tabular datasets like employee records, handles both numerical and categorical features well, and is less sensitive to noise in the data. It also performs reliably without extensive parameter tuning, making it a good baseline model for attrition prediction.

- **XGBoost** is a boosting-based algorithm that builds decision trees one after another, where each new tree focuses on correcting the mistakes made by previous ones. This allows the model to learn more complex patterns in the data. We selected XGBoost because it often achieves high accuracy on structured datasets and can capture subtle relationships between employee characteristics and attrition. It is especially useful when multiple factors together influence the target variable, which is common in employee retention problems.

Using these two techniques allows us to compare a simpler and more stable ensemble method (Random Forest) with a more powerful and optimized boosting method (XGBoost). This comparison helps determine which approach is more effective and practical for predicting employee attrition in this dataset.

## Dataset Description

The ***Employee Attrition & Retention Analytics*** Dataset, obtained from Kaggle, will be used in this project. The data will be split into training and testing sets to train the models and assess their performance.

The dataset is available at:
https://www.kaggle.com/datasets/ajinkyachintawar/employee-attrition-and-retention-analytics-dataset.

The dataset contains historical employee data from an organization, including features such as age, job role, department, salary level, job satisfaction, work-life balance, overtime status, years of experience, along with other demographic, professional, and satisfaction-related attributes. The target variable, *Attrition*, indicates whether an employee left the company (Yes) or remained with the organization (No). The dataset includes both numerical and categorical attributes, making it suitable for tree-based ensemble learning methods such as Random Forest and XGBoost.

### Evaluation Plan

Both models will be evaluated using standard classification performance metrics, including accuracy, precision, recall and F1-score. The comparative evaluation of these two techniques will determine which model provides better support for an intelligent employee attrition prediction agent and offers more reliable insights for organizational decision-making.